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# ACCRETION OF MATTER BY CONDENSED OBJECTS

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**Abstract.** The equations of motion for steady-state spherical symmetric flow of matter into or out of a condensed object (e.g. neutron stars, 'black holes', etc.) are displayed and solved for simple polytropic gases. It appears that infalling matter may be heated as hot as $10^{12}$ K and that X-ray luminosities of the order of $10^{37}$ erg s$^{-1}$ could result. The two fluid (electrons and ions separately) approach is also examined and it is shown that electrostatic fields of the order of $10^5$ V m$^{-1}$ are required near the surface of the object. Such fields are not strong enough to significantly modify the space-time metric.

## 1. Introduction

A collapsed object surrounded by the thin interstellar plasma is expected to capture particles that pass within a certain distance. The result must be a diffusion or flow of the plasma towards the object. We will assume that a continuum description of the plasma is an adequate approximation (e.g. we could postulate a weak magnetic field to couple the particles together, but otherwise ignore such fields). The collapsed object is represented by a Schwarzschild singularity and corrections to the metric from the external plasma itself are dropped. The nonrelativistic version of this model has been treated by Bondi (1951).

The basic equations of motion are conservation of mass flux,

$$J^k{}_{;k} = 0\,, \tag{1}$$

and energy flux,

$$T^k_{0\,;k} = 0\,, \tag{2}$$

where the semicolons denote the covariant derivatives. These equations have quite simple forms if steady state conditions and spherical symmetry are assumed, becoming

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(J^1\sqrt{-g}\right) = 0 \tag{3}$$

and

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(T^1_0\sqrt{-g}\right) = 0\,. \tag{4}$$

Integration is trivial, and for a perfect fluid we can write

$$\varrho U^1\sqrt{-g} = C_1 \tag{5}$$

and

$$(P+\mu)\,U_0U^1\sqrt{-g} = C_2\,, \tag{6}$$

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where $C_1$ and $C_2$ are constants, $\varrho$ is the proper density (i.e. that seen moving with the flow), $P$ is the proper pressure, and $\mu$ is the total proper energy density. For the Schwarzschild metric we have

$$-\mathrm{d}s^2 = e^{\lambda}\,\mathrm{d}r^2 + r^2\,(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\varphi^2) - c^2 e^{\nu}\,\mathrm{d}t^2\,, \tag{7}$$

and the remaining terms in (6) are $U^1 = \mathrm{d}r/\mathrm{d}s$, $U_0 = e^{\nu}\mathrm{d}t/\mathrm{d}s$, and $-g = c^2 r^4 \sin^2\theta\, e^{\nu+\lambda}$, as usual. Given an equation of state that relates $P$ and $\mu$ to $\varrho$, we have two Equations (5) and (6) in two unknowns ($\varrho$ and $U^1$) at each value of $r$. For the Schwarzschild exterior metric, $e^{\nu} = e^{-\lambda} = 1 - 2m/r$ with $m = GM/c^2$ and the conservation equations become

$$\varrho u r^2 = C_1 \tag{8}$$

and

$$\left(\frac{P+\mu}{\varrho}\right)^2 \left(1 - \frac{2m}{r} + u^2\right) = (C_2/C_1)^2 = C_3\,, \tag{9}$$

where $u = U^1/c$ and (6) has been divided by (5) and squared to yield (9). The total energy density is given by

$$\mu = \varrho c^2 + \varepsilon\,, \tag{10}$$

where $\varepsilon$ is the internal (thermal) energy. In the nonrelativistic limit $C_3 \to 1$ and the usual solar wind equations are obtained by subtracting off unity from the left hand side of (9) and dropping second-order terms.

These equations are characterized by a critical point, as is usual for hydrodynamic flow systems. Differentiation (8) and (9) and elimination of $\mathrm{d}\varrho$ gives

$$\frac{\mathrm{d}u}{u}\left[2V^2 - \frac{m}{r\left(1 - \frac{2m}{r} + u^2\right)}\right] + \frac{\mathrm{d}r}{r}\left[V^2 - \frac{u^2}{\left(1 - \frac{2m}{r} + u^2\right)}\right] = 0\,, \tag{11}$$

with

$$V^2 \equiv \frac{\mathrm{d}\ln(P+\mu)}{\mathrm{d}\ln\varrho} - 1\,. \tag{12}$$

It is evident that if one or the other of the bracketted factors in (11) vanishes one has a turn-around point, and the solutions are double-valued in either $r$ or $u$. Only solutions that pass through a critical point correspond to material falling into (or flowing out of) the object with monotonically increasing velocity along the particle trajectory. The critical point is located where both bracketted factors in (11) vanish, thus

$$u_c^2 = m/2r_c \tag{13}$$

and

$$V_c^2 = u_c^2/(1 - 3u_c^2)\,. \tag{14}$$

No physically acceptable solution exists for $V_c^2$ if $u_c^2 > \frac{1}{3}$ or $r_c < 6m$, thus the critical point is always outside of a fully collapsed object (surface at $r \to 2m$).

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## 2. Simple Polytropic Solutions

For the equation of state

$$P = K\varrho^{\gamma} \tag{15}$$

we have, for constant $\gamma$,

$$P + \varepsilon = \frac{\gamma}{\gamma - 1} P, \tag{16}$$

and defining

$$T \equiv P/\varrho \tag{17}$$

(for protons, these temperature units are $m_p c^2/k = 1.09 \times 10^{13}$ K, or essentially $T_{13}$ in one system of notation) we find

$$T^n u r^2 = C_4, \tag{18}$$

$$[1 + (1 + n)\, T]^2 \left(1 - \frac{2m}{r} + u^2\right) = C_5, \tag{19}$$

and

$$V^2 = \frac{(1 + n)\, T}{n[1 + (1 + n)\, T]}, \tag{20}$$

where

$$n = 1/(\gamma - 1), \tag{21}$$

which is the usual relation between polytropic index ($n$) and the Poisson adiabatic index ($\gamma$). There are, of course, a number of other equations of state that might be argued to be more appropriate, but we will illustrate the properties of the flow equation for $\gamma = \frac{4}{3}$ (relativistic, conserved, elementary particles – e.g. hot electrons) and $\gamma = \frac{5}{3}$ (nonrelativistic, conserved, elementary particles – e.g. cool protons). There is no reason that more sophisticated equations of state can not be employed except that the analysis would be complicated algebraically and thereby dominated by numerical details. Such analysis would nevertheless be straightforward. Since the gas is likely to be ionized, it is interesting to see what role the resultant electric field plays. Were the ions and electrons uncoupled, they would in general have different flow characteristics which would in itself lead to relatively large excursions from local charge neutrality. These considerations are examined in the Appendix, where it is shown that the fundamental Equations (8) and (9) remain valid.

Figure 1 shows the behavior of solutions passing through the critical point – there are two branches, one corresponding to low velocities at infinity and high velocities as $r \to 0$ and the other, vice versa. The sign of $u$ is not restricted, but for likely astrophysical circumstances, these two branches represent the infall and outflow of matter respectively. The above interpretation is in no way essential to the analysis, however.

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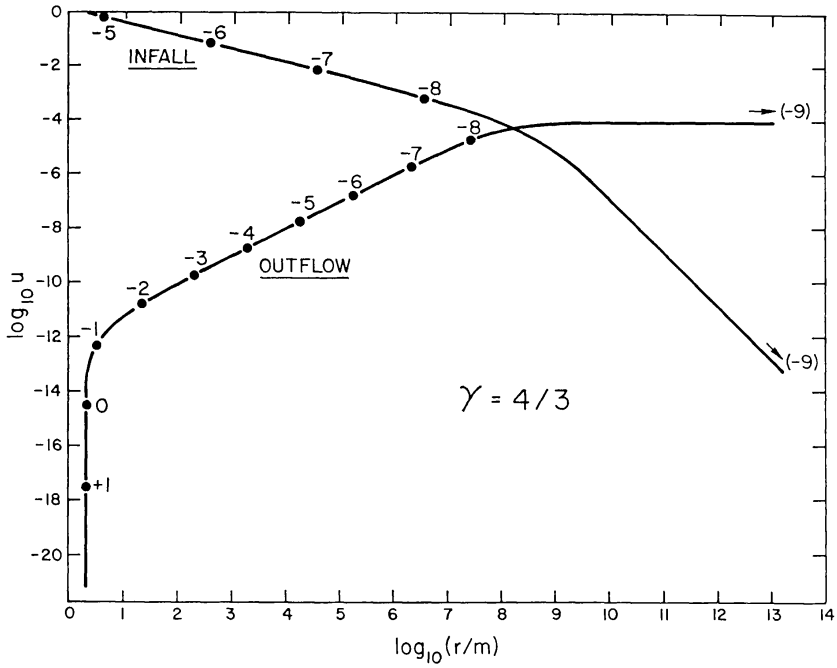


Fig. 1. Logarithmic plot of $u$ (velocity in units of $c$) verses distance (in units of gravitational radius, $m \equiv GM/c^2$) for a $\gamma = \frac{4}{3}$ gas having an asymptotic temperature at infinity of $10^{-9}\ m_p c^2/k$ or about $10^4$ K and falling into a collapsed object. The companion curve labeled 'outflow' corresponds to the same mass and energy flux moving *away* from the star, in which case the temperature approaches zero but the velocity is asymptotic to $10^{-4}\ c$. The gas temperature at different distances from the object is given as $\log_{10}(kT/m_p c^2)$ at the indicated points along the curve.

Given $r_c$ and $u_c$ we then know $T_c$, hence $C_4$ and $C_5$, hence as $r \to \infty$ we can determine $T_\infty$ and $u_\infty$ for the two branches as well as $T_s$ and $u_s$ at $r = r_s$ ($\geqslant 2m$). Thereby the surface conditions can be expressed in terms of the conditions at infinity, and the conditions at the critical point are given as an intermediate step in the calculation.

CASE I. $\gamma = \frac{4}{3}$ ($n = 3$)

At infinity we expect for the 'infall' branch that $u_\infty = 0$ and $T_\infty$ is some nonzero value, the temperature of the local interstellar medium. Therefore $C_4$ is indeterminate and $C_5$ is nearly unity in our notation (e.g. $T_\infty = 10^{-9}$ at $10^4$K). But

$$C_5 \approx 1 + 8T_\infty \tag{22}$$

and

$$C_5 \approx (1 - 3u_c^2)^3/(1 - 6u_c^2)^2 \approx 1 + 3u_c^2 . \tag{23}$$

Furthermore,

$$4T_c = 3u_c^2/(1 - 6u_c^2) \approx 3u_c^2 ; \tag{24}$$

thus leading to

$$T_c = 2T_\infty . \tag{25}$$

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Then,

$$C_4 = T_c^3 u_c r_c^2 = \frac{3\sqrt{6}}{16} T_\infty^{3/2} m^2 , \tag{26}$$

and the constants are all determined in terms of $T_\infty$. At the observable surface of the object, we assume as a first approximation that $T_s$ will probably be much less than unity, then $u^2 \approx 2m/r_s$ and therefore from (26)

$$T_s = \frac{\sqrt{6}}{4} T_\infty^{1/2} (m/r_s)^{2/3} . \tag{27}$$

For $T_\infty = 10^{-9}$ ($\approx 10^4$ K), $r_c \approx 2$ AU, $u_c \approx 14$ km s$^{-1}$, and $T_s = 3 \times 10^7$ K at $r_s = 2m$. The density does not appear explicitly, but is given from

$$\varrho/\varrho_\infty = (T/T_\infty)^3 , \tag{28}$$

thus a compression factor of $3 \times 10^{10}$ would be achieved in the above example.

CASE II. $\gamma = \frac{5}{3}$ ($n = \frac{3}{2}$)

For this case

$$\tfrac{5}{3} T_c = 3u_c^2/(2 - 9u_c^2) \approx \tfrac{3}{2} u_c^2 \tag{29}$$

thus

$$C_4 = T_c^{3/2} u_c r_c = 3\sqrt{15}\, m^2/100 = (0.116 \ldots)\, m^2 , \tag{30}$$

and is independent of the external conditions. Thus the temperature at the surface is

$$T_s = 0.3\, (m/r_s)^{4/3} . \tag{31}$$

Proton temperatures of $10^{12}$ K would be achieved provided that the above equation of state were still valid up to the critical point. The critical point is determined as before, since

$$C_5 \approx 1 + 5T_\infty \tag{32}$$

and

$$C_5 = 4(1 - 3u_c^2)^3/(2 - 9u_c^2)^2 \approx 1 + 27u_c^4/4 \tag{33}$$

or

$$u_c^4 = 20T_\infty/27 . \tag{34}$$

For $T_\infty = 10^{-9}$ as before, the critical point parameters are $u_c = 1.6 \times 10^3$ km s$^{-1}$, $r_c = 2.7 \times 10^4$ km$^{-1}$, and $T_c = 1.8 \times 10^8$ K. The compression factor is $(T/T_\infty)^{3/2}$ and has a value of $1.0 \times 10^{13}$ at the surface in the above example.

## 3. Energy Flux

We can put an upper limit to the intrinsic gas luminosity by assuming that the entire internal energy radiated outwards just prior to crossing the singularity at $r = 2m$. If the infall is onto a stationary surface, however, one also has available the flow energy,

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but at larger radius, thus a lesser gravitational potential. These two luminosities are, for the $\gamma = \frac{5}{3}$ case,

$$L_{bh} \leqslant (\tfrac{3}{2} T_s m_p c^2)\,(4\pi C_4)\,(n_\infty T_\infty^{-3/2}) \tag{35}$$

$$\approx (8 \times 10^{36}\ \mathrm{erg\ s^{-1}})\, n_\infty\,(\mathrm{cm^{-3}})\, T_\infty\,(\mathrm{K})^{-3/2} \tag{36}$$

for a black hole with $r_s = 3$ km. For comparison, the gravitational energy release rate alone for infall onto a neutron star with $r = 10$ km would be essentially the same: namely,

$$L_{ns} \approx (1.1)\, L_{bh}(\max). \tag{37}$$

## 4. X-ray Sources?

One might speculate that some of the X-ray sources result from matter falling into a condensed object. See for example Shvartsman (1970). In the $\gamma = \frac{5}{3}$ case it would appear that adequate luminosities would result, either if the gas radiates the bulk of its internal energy just before falling into a central 'black hole' or if the total flow energy is liberated by impact with the surface of a neutron star. One must, however, somehow transfer away most of the angular momentum that the infalling gas had relative to the center of mass. It seems physically plausible that the effect of such angular momentum would be to choke down the inflow rate. For example, even when magnetic torques are included (see Figure 1 in Michel (1969)) one finds that the 'infall' solutions terminate at finite distances from the origin in analogy with the minimum approach distance of a single particle trajectory having nonzero initial angular momentum. On the other hand, if the infalling gas originated from the same star that produced the condensed object as well, then a low angular momentum density might be contemplated for that gas.

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## Appendix: Charge Separation Fields

In the text, the implicit assumption was made that all components of the gas were coupled together (e.g. by collisions). Here we examine the role of electrostatic fields in coupling the electrons and ions together in an ionized infalling gas.

We will treat the ions and electrons independently, in which case Equation (4) becomes

$$\frac{\mathrm{d}}{\mathrm{d}r}\,(T_0^1 \sqrt{-g})_s = F_{01} j_s / c \quad (s = e \quad \text{or} \quad i). \tag{A1}$$

If charged particles are moving along electric field lines, their energy flux must change as a function of position, and the right-hand side of (A1) is nonzero owing to the work

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being done by a radial current $j_s$ in the (radial) field $F_{01}g$ where $s$ indicates either the electron ($e$) or ion ($i$) components.

For steady state conditions, however, we also have

$$\sum j_s = j_e + j_i = 0\,, \tag{A2}$$

otherwise the object would be gaining or losing charge. It follows immediately that

$$\frac{\mathrm{d}}{\mathrm{d}r} \sum (T_0^1 \sqrt{-g})_s = 0\,. \tag{A3}$$

Since

$$T_0^1 = (P + \mu)\left(1 - \frac{2m}{r} + u^2\right)^{1/2} uc\,, \tag{A4}$$

we can simply add together the pressures and internal energies, provided that $u_e = u_i$, which, from (A2) and the definition

$$j_s = 4\pi q_s n_s u_s c \sqrt{-g}\,, \tag{A5}$$

is equivalent to assuming $n_e = n_i$ for singly charged ions ($g_e = -g_i$), i.e. microscopic charge neutrality. Neutrality can be established by calculating the resultant fields and showing that the required space charge is negligible compared to that available. The anticipated result is then that

$$\begin{aligned} P &= \sum P_s\,, \\ \mu &= \sum \mu_s\,, \end{aligned} \tag{A6}$$

and that $u$ and $n$ are independent of $s$.

Subtracting the two Equations (A1) gives

$$\frac{\mathrm{d}}{\mathrm{d}r}\left([T_0^1(e) - T_0^1(i)]\sqrt{-g}\right) = 2F_{01} j_e \sqrt{-g}/c\,, \tag{A7}$$

and for equal pressures (an assumption which we will drop)

$$T_0^1(e) - T_0^1(i) = (m_e - m_i)\, n\, (e^{\nu} + u^2)^{1/2} uc\,. \tag{A8}$$

If we use (A5), the combination $nu\sqrt{-g}$ is recognized to be a constant proportional to $j_e$, giving thereby the result

$$F_{01} = \frac{(m_i = m_e)\, c}{8\pi e} \frac{\mathrm{d}}{\mathrm{d}r}\left(1 - \frac{2m}{r} + u^2\right)^{1/2}. \tag{A9}$$

For example, if the particles are all in free fall (e.g. a zero temperature gas), the electrons and ions fall exactly together and no electric field is required. For free fall, the equations of motion are

$$1 - \frac{2m}{r} + u^2 = \text{const} \tag{A10}$$

and from (A9) we have $F_{01} = 0$ as expected.

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In general (unequal pressures, etc.) we have

$$F_{01} = \frac{c}{8\pi e}\frac{\mathrm{d}}{\mathrm{d}r}\left[\left(\frac{P_i + \mu_i - P_e - \mu_e}{n}\right)\left(1 - \frac{2m}{r} + u^2\right)^{1/2}\right]. \tag{A11}$$

One can estimate roughly the order of magnitude of the electric field in (A11), since the energy change given to the ions is of the order of the rest mass ($\approx 10^9$ eV for protons) over a distance of a stellar radius ($\approx 10^3$ m for a collapsed object), or about $10^6$ V m$^{-1}$. Such a field corresponds to a net charge of $10^2$ C, or about one excess charge per $10^{36}$ nucleons.

The space-time curvature about a collapsed object is usually assumed to be the Schwarzschild exterior metric (7) with

$$e^{-\lambda} = 1 - \frac{2GM}{rc^2} = e^{\nu}. \tag{A12}$$

If the object also possesses a charge, however, the metric coefficient becomes (c.f. Tolman, 1934)

$$e^{-\lambda} = 1 - \frac{2GM}{rc} + \frac{GO^2}{r^2c^4} = e^{\nu}. \tag{A13}$$

Among the interesting properties of the latter metric is the lack of a singularity provided that

$$Q^2 > GM^2. \tag{A14}$$

The electric charge given from (A11) is, however, very much less than that required by (A14). One requires a charge excess of one in $10^{18}$ to seriously modify the spacetime metric near a star, thus such a field is unimportant in that sense. Large fields could arise only if the pressure gradient were extremely large near the surface. From (9), it is clear that large gradients occur only if $u \to 0$ as $r \to 2m$. Such behavior corresponds to the stellar wind type solutions wherein an extremely hot gas expanded away from the surface of a collapsed object. Formally, the temperature is infinite at $r = 2m$, but the metric is no longer the same and one must do a self-consistent calculation.

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